

Question

The chemical formula Na_3NO_3 represents a cubic crystal in which all sodium atoms exist as Na^+ ions and are located at the face centers of the unit cell. In addition to Na^+ , there are two other ions in the unit cell. If the one with the larger mass is labeled **A** and the smaller one is labeled **B**, what are the charge ratio of **A** to **B** and the difference in the number of nearest-neighbor Na^+ ions between **A** and **B** in the crystal?

A. 0.5; 6

B. 0.5; 3

C. 0.5; -3

D. 0.5; -6

E. 2; 6

F. 2; 3

G. 2; -3

H. 2; -6

Answer

Correct Answer: **A**

Detailed Explanation

A unit cell contains 3 face-centered Na^+ ions and two other ions. From the chemical formula Na_3NO_3 , it is known that these two ions collectively contain 1 N and 3 O atoms, with a total charge of -3. Among common ions, only two combinations correspond to this:

1. NO^- and O_2^{2-}
2. NO_2^- and O^{2-}

Considering the strong oxidizing property of O_2^{2-} and the reducing property of NO^- , the second scenario is more reasonable. The molecular weight of NO_2^- is approximately 46, and that of O^{2-} is approximately 16. Therefore, **A** is NO_2^- , and **B** is O^{2-} . Their charges are -1 and -2, respectively, giving a ratio of 0.5.

CHECKPOINT

3 PTS

A is NO_2^- , **B** is O^{2-}

CHECKPOINT

1 PTS

The charge ratio of **A** to **B** is 0.5

Next, consider the crystal structure. The Na^+ ions occupy face-centered positions, generating within a single unit cell one vertex and one body-centered position, totaling two voids that satisfy cubic symmetry, which can be

allocated to the two anions.

Let the edge length of the unit cell be a . The distance between a vertex and an adjacent Na^+ is $\frac{\sqrt{2}}{2}a$, with 4 Na^+ ions along the xy , yz , and xz plane directions each, totaling 12 nearest neighbors. The distance between the body center and an adjacent Na^+ is $0.5a$, with 2 Na^+ ions along the x , y , and z -axis directions each, totaling 6 nearest neighbors.

CHECKPOINT

2 PTS

The vertex has 12 nearest-neighbor Na^+ ions, and the body center has 6 nearest-neighbor Na^+ ions

The vertex has a larger distance to Na^+ and is thus suitable for the larger, less-charged anion. Therefore, **A** should occupy the vertex, and **B** should occupy the body center. The numbers of nearest-neighbor Na^+ ions for **A** and **B** are 12 and 6, respectively, giving a difference of 6. Option A corresponds to this.

CHECKPOINT

1 PTS

A is located at the vertex, **B** is located at the body center

CHECKPOINT

1 PTS

The difference in the number of nearest-neighbor Na^+ ions between **A** and **B** is 6